REVIEW

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AI-enabled drug and molecular discovery: computational methods, platforms, and translational horizons

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Abstract
The integration of artificial intelligence (AI) with bioinformatics has initiated a transformative shift in drug discovery, redefining how pharmaceutical research and development are conducted. This review examines both the current state and emerging prospects of AI-driven strategies across the drug discovery pipeline, from target identification and molecular design to clinical applications. Advances in machine learning, deep learning, graph neural networks, transformers, foundation models, and quantum computing have shown remarkable potential in overcoming long-standing bottlenecks of conventional drug development, which on average requires 12.5 years and over $2 billion to bring a single drug to market. The AI in pharmaceuticals market, valued at $1.8 billion in 2023, is projected to reach $13.1 billion by 2030, reflecting a compound annual growth rate of 18.8%. Landmark breakthroughs such as AlphaFold, which has generated over 200 million predicted protein structures and more than 20,000 citations, have established new paradigms in structural biology and drug design, while emerging tools like RFdiffusion and AlphaFold3 further extend capabilities into *de novo* protein design and multi-omics integration. AI-enabled workflows have demonstrated the ability to compress discovery timelines from five years to as little as 12–18 months and reduce costs by up to 40%. Despite these advances, significant challenges remain, including issues of data quality and bias, model transparency and interpretability, computational resource demands, and ethical concerns regarding privacy and algorithmic fairness. This review synthesizes findings from recent studies to provide a roadmap for the responsible and reproducible integration of AI into bioinformatics and drug discovery, while outlining critical gaps and future research priorities essential for clinical translation.

Keywords Artificial intelligence, Bioinformatics, Drug discovery, Drug design

1 Introduction

Drug discovery has a long history, originating in ancient civilizations where remedies derived from natural sources were employed for physical and spiritual healing. Modern drug discovery, which emerged in the early twentieth century, typically begins with academic research aimed at identifying disease-related macromolecules, pathways, or mechanisms during the pre-discovery stage. The process involves multiple steps target identification, hit discovery, lead optimization, preclinical evaluation, and clinical trials that demand multidisciplinary expertise and advanced technologies to identify and validate therapeutic candidates [1, 2]. Despite advances in experimental technologies, high attrition rates and escalating development costs continue to pose major challenges.

In recent years, artificial intelligence (AI) has emerged as a transformative force in pharmaceutical research. AI methodologies including machine learning (ML), deep learning (DL), reinforcement learning (RL), and natural language processing (NLP) enable the rapid analysis of vast chemical and biological datasets, prediction of molecular properties, hypothesis generation, and automated molecule design [3, 4]. These approaches offer the potential to accelerate multiple stages of the drug discovery pipeline, from virtual screening and structural modeling to toxicity prediction and clinical trial optimization.

A wide range of AI platforms and tools exemplify this transformation. IBM Watson and Benevolent AI assist in data analysis and target identification; DeepMind's AlphaFold has revolutionized protein structure prediction; Atom wise and Schrödinger employ deep learning and computational chemistry for molecular interaction prediction; OpenAI's GPT models support hypothesis generation and text mining; while NVIDIA Clara provides infrastructure for genomics and medical imaging [3, 5–7]. Together, these platforms enhance predictive accuracy, streamline experimental workflows, and accelerate the overall pace of drug discovery. Neural networks contribute to predicting molecular bioactivity [6], deep learning advances structural biology and biomedical data analysis [5], and reinforcement learning improves molecular design and treatment optimization [8].

The purpose of this paper is to provide a comprehensive narrative overview of how AI-driven methods, computational platforms, and translational applications are reshaping drug and molecule discovery. It is important to clarify that this work is a narrative review, synthesizing key developments and trends from the literature to offer a broad perspective on the field. Unlike a systematic review, it does not follow strict, protocol-driven literature selection criteria. Instead, it surveys core AI methodologies including machine learning, deep learning, reinforcement learning, and natural language processing alongside the databases and tools that enable their deployment, with the aim of illustrating the transformative impact and current landscape of AI in pharmacology.

This overview highlights the application of these technologies across the discovery pipeline, from hit identification and *de novo* molecular design to ADMET prediction, target validation, and drug repurposing. It also explores synergies with complementary innovations such as CRISPR, quantum computing, and high-throughput screening, illustrated through case studies from academia and industry. This paper discusses current challenges, ethical considerations, and future directions, offering a perspective on responsibly integrating AI into next-generation drug discovery to maximize translational impact. The growing economic impact and performance metrics of AI in drug discovery are summarized in Table 1.

2 Evolution of computational biology

The landscape of biological research has undergone a profound transformation over the past decades, driven by the exponential growth of biological data and computational capabilities [9]. Traditional approaches to understanding biological systems, once limited by experimental throughput and analytical capacity, have evolved into sophisticated computational frameworks capable of processing vast multi-dimensional datasets [10]. This evolution has been particularly pronounced in drug discovery, where the integration of computational methods has become essential for navigating the complexity of modern pharmaceutical research [11]. The emergence of high-throughput technologies, including next-generation sequencing, proteomics platforms, and metabolomics approaches, has generated unprecedented volumes of biological data [12]. This data explosion, often characterized as biology's transition to becoming the paradigmatic "big data" producer, has necessitated the development of advanced computational methodologies capable of extracting meaningful insights from these complex datasets [13].

2.1 Integration of AI in bioinformatics

Artificial intelligence has arisen as a transformative force in bioinformatics, offering sophisticated approaches to data analysis, pattern recognition, and predictive modelling that were previously unattainable [14]. The integration of AI techniques, particularly machine learning and deep learning, has enabled researchers to move beyond descriptive analyses toward predictive modelling capabilities that can anticipate biological outcomes and guide experimental design [15].

Machine learning applications in bioinformatics have demonstrated remarkable success across diverse domains, including gene identification, protein function prediction, and the discovery of functional modules within biological networks [16]. Deep learning architectures, such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs), have proven particularly effective in handling the complex, hierarchical structures inherent in biological data [17].

2.2 Significance in drug discovery

The pharmaceutical industry faces mounting pressure to increase efficiency and reduce the substantial costs and time requirements associated with bringing new therapies to

Table 1 Market statistics and performance metrics

Metric	Value	Source
Global AI Drug Discovery Market 2023	\$1.39–1.86 billion	Markets and Markets, Grand View Research
Projected Market 2030	\$6.89–20.30 billion	Grand View Research
CAGR (2024–2030)	29.7–29.9%	Multiple industry reports
Traditional Drug Development Time	14.6 years	Industry standard
AI-Accelerated Timeline	12–18 months	Exscientia, industry reports
Traditional Success Rate	10%	Pharmaceutical industry average
Cost Reduction Potential	Up to 40%	Industry estimates
AlphaFold Database Entries	> 200 million structures	AlphaFold Database
RFdiffusion Success Rate	23/25 benchmark problems	Nature publication
AtomNet Library Size	3 + trillion compounds	Atomwise platform

market [18]. Traditional drug discovery processes, characterized by high failure rates and extended development timelines, have created an urgent need for innovative approaches that can improve success rates while reducing resource requirements (Table 3).

AI-driven methodologies have demonstrated significant potential in addressing these challenges through applications spanning the entire drug discovery pipeline [19]. From target identification and validation to clinical trial optimization, AI technologies are enabling more rational, efficient, and effective approaches to pharmaceutical research (Fig. 1) [20]. The technology has shown particular promise in virtual screening, where AI-powered platforms can rapidly evaluate vast compound libraries to identify promising drug candidates.

3 AI technologies in drug discovery

Artificial intelligence encompasses a broad spectrum of computational methodologies that enable machines to learn from data, recognize complex patterns, and make predictive or generative decisions [21]. In drug discovery, AI models are broadly categorized into Machine Learning (ML), Reinforcement Learning (RL), Deep Learning (DL), Generative AI, and Natural Language Processing (NLP). Each paradigm contributes uniquely across different stages of molecular design, virtual screening, target identification, and property prediction [22]. An overview of these core AI technologies, their applications, advantages, and limitations is summarized in Table 2.

3.1 Machine learning (ML)

Machine Learning (ML) provides a powerful framework for building predictive models from structured datasets, making it indispensable for key tasks in drug discovery such as quantitative structure-activity relationship (QSAR) modeling, prediction of absorption, distribution, metabolism, excretion, and toxicity (ADMET) properties, and biomarker identification [23]. Its applications extend to enhancing target validation, prognostic biomarker discovery, and digital pathology analysis. However, the effectiveness of ML is

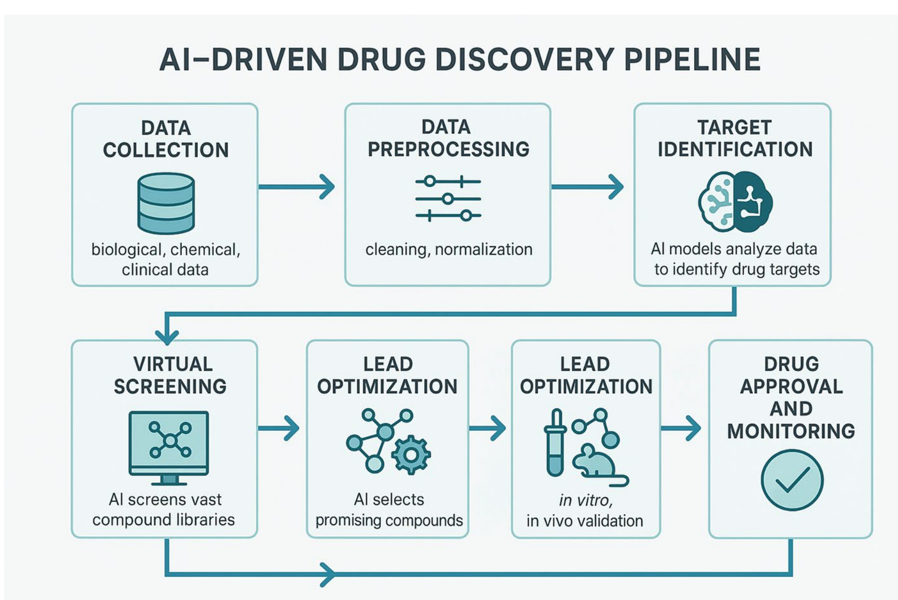

Fig. 1 Flow chart illustrating stages of AI-driven drug discovery

Table 2 **AI** technologies in bioinformatics applications

Technology	Applications	Advantages	Limitations	Example Tools/Models
Machine Learning (ML)	QSAR modeling, biomarker identification	Interpretable, fast training	Limited complexity handling	Random Forest (RF), Support Vector Machine (SVM), k-Nearest Neighbors (kNN)
Deep Learning (CNNs)	Molecular screening, protein structure	High accuracy for image-like data	Black box, computationally intensive	AlphaFold, AtomNet, DeepChem
Deep Learning (RNNs)	Sequence analysis, time-series	Sequential data processing	Vanishing gradient problems	LSTMs, GRUs for SMILES-based generation
Generative Adversarial Networks (GANs)	Novel molecule generation	Novel structure generation	Training instability	GENTRL, ORGAN
Natural Language Processing (NLP)	Literature mining, knowledge extraction	Automated knowledge extraction	Domain-specific tuning needed	BioBERT, SciBERT, ChemDataExtractor
Quantum Machine Learning	Molecular optimization, quantum effects	Quantum advantage potential	Hardware limitations	Variational Quantum Eigensolver (VQE)
Graph Neural Networks	Molecular property prediction	Molecular graph understanding	Limited interpretability	ChemProp, Message Passing Neural Networks (MPNNs)
Transformer Models	Protein language modeling	Attention mechanisms	Large computational requirements	ProtBERT, MolBERT, MegaMolBART

governed by critical factors including data quality, model interpretability, and the consistency of results, which can limit its adoption.

The complexity of drug discovery requires analytical capabilities that can integrate large, diverse datasets to uncover patterns and relationships not easily discernible through conventional techniques. A primary challenge is the dependency on substantial volumes of high-quality data; models often struggle with data sparsity and lack transparency in their decision-making processes [24]. Ongoing research is addressing these limitations through explainable AI (XAI) approaches, advanced feature engineering, and hybrid modeling techniques, which are expanding the practical impact of ML in the field [25].

3.1.1 Supervised vs. unsupervised learning

The selection of an ML paradigm is dictated by the nature of the available data. Supervised learning relies on labeled datasets to train models for prediction and is the cornerstone of tasks such as classifying compounds as active/inactive or predicting continuous properties like binding affinity. Its two primary tasks are classification (predicting categorical labels) and regression (predicting continuous values). In contrast, unsupervised learning identifies inherent patterns, structures, and relationships within unlabeled data. Key techniques include clustering algorithms (e.g., k-means, hierarchical clustering) for grouping similar compounds or patient subtypes, and dimensionality reduction methods like Principal Component Analysis (PCA) and t-distributed Stochastic Neighbor Embedding (t-SNE) for visualizing and interpreting high-dimensional biological data [26]. These methods are pivotal for exploratory analysis, such as analyzing high-throughput screening data or stratifying patients based on multi-omics profiles [27].

These paradigms are not mutually exclusive. Semi-supervised learning hybrid approaches leverage both labeled and unlabeled data, making them valuable for problems where labeled data is scarce [28]. Together, supervised and unsupervised learning provide complementary strengths for extracting knowledge from complex biomedical datasets.

3.1.2 Supervised machine learning algorithms

- **Random Forest (RF)**: This ensemble algorithm constructs multiple decision trees to improve predictive accuracy and control over-fitting. It is widely used for both classification (e.g., predicting bioactive compounds) and regression tasks (e.g., modeling drug response values like IC50) [29, 30]. Its robustness and ability to handle high-dimensional data make it a staple in chemoinformatics.
- **Support Vector Machines (SVM)**: SVMs are highly configurable algorithms effective for classification and regression. They excel in tasks like compound categorization and molecular property forecasting by finding optimal hyperplanes to separate data classes [31]. Despite the rise of deep learning, SVMs remain competitive due to their strong generalization performance and relative interpretability, as demonstrated in predicting bioactivity for targets like dihydrofolate reductase [32, 33].
- **k-Nearest Neighbors (kNN)**: As a simple, distance-based algorithm, kNN classifies data points based on the majority class of their 'k' nearest neighbors in the feature space. Its interpretability makes it useful in chemoinformatics for tasks like bioactivity prediction. Its performance is often enhanced through optimization techniques for feature selection and by using topological molecular descriptors [34–36].

3.1.3 Unsupervised machine learning algorithms

Unsupervised learning is essential for exploratory data analysis and hypothesis generation in the early drug discovery pipeline. It serves critical functions such as:

- **Clustering**: Grouping compounds with similar structural or physicochemical properties to analyze chemical libraries and identify novel scaffolds for lead generation.
- **Dimensionality Reduction**: Techniques like PCA and t-SNE are indispensable for visualizing and interpreting high-dimensional data (e.g., transcriptomic or proteomic profiles), revealing underlying patterns and patient subtypes for precision medicine.

By elucidating the intrinsic organization of complex biological and chemical data, unsupervised learning provides a foundational analysis that guides subsequent experimental design and supervised modeling (Fig. 2).

3.2 Reinforcement learning (RL)

RL for Structural Evolution (ReLeaSE) is a computational advance for developing molecules with explicit quality through RL and deep neural networks. It combines generative and predictive models for focused chemical libraries creation. Generative models rely on stack-augmented memory network, While, predictive models anticipate certain qualities. This approach can be employed to develop chemical libraries focusing on structural complexity, physical characteristics, or inhibitory effects on Janus protein kinase [48].

RL techniques have extended into wider pharmacological applications, contributing potential strategies for providing personalized treatments for neurological disorders and

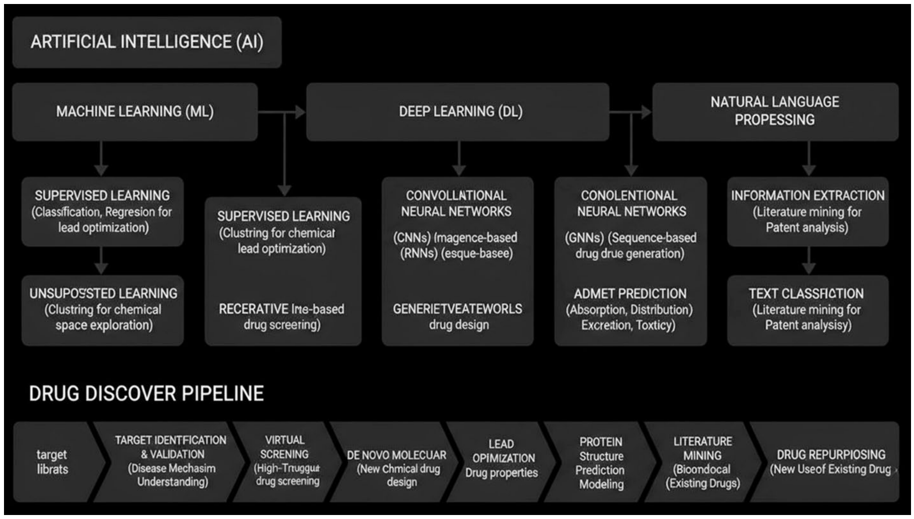

Fig. 2 Schematic Overview of AI/ML Methodologies in Drug Discovery

inflammatory diseases. Recent studies underline advanced RL methods in drug design, highlighting on key challenges and potential solutions in personalized medicine [49].

Moreover, these methodologies are progressively more effective in understanding difficult disease-specific drug development. Tan et al. [50] clarifies that the investigated method supports tailored treatments for complex diseases, addressing barriers and enabling focused medication development. Consequently, RL suggest possible avenues for *de novo* drug design and personalized therapy, despite the fact that its practical application is still developing.

3.2.1 Molecule generation and optimization

Molecular optimization plays a chief role in drug discovery, especially improving the chemical properties. An optimization technique includes methods based on molecular mapping, molecular distribution matching, and assisted search strategies. Current challenges in molecular optimization involve interpreting, multidimensional refinement, and model generalization. Recent time researches underscore the difficulties and prospects in molecular optimization using artificial intelligence for drug discovery [51].

Among these AI-based strategies, generative models have developed into an effective yet complex tool for molecular refinement. Additionally, they are commonly employed to improve the characteristics and biological impacts of chemical compounds [52].

These innovations in generative modeling represent the broader impact of AI across the drug discovery. Recent studies in molecule generation and optimization have significantly improved drug discovery and development. These advancements have altered the sector by combining multiple disciplines and dealing issues in data and technical dimensions. These applications of Molecular optimization are currently employed in multiple areas, speeding up drug discovery and development [53]. Consequently, *an* AI-driven molecular generation and optimization accelerate drug discovery by improving compound properties and addressing intricate design challenges.

3.2.2 Game-theoretic approaches in Docking and protein-ligand interaction

These game-theoretic strategies are becoming more interconnected with digital platforms to simulate and optimize biological interactions. Deep cyberspace, a rapidly evolving computing conception, integrates real-world and computer-generated 3D environments. Strategically assisting advancement through multistage games and opponent exploitation can aid in therapies, cell reprogramming, and synthetic biology. This method holds potential uses in therapy, cell reprogramming, and synthetic biology [54].

In recent times, consumers can operate virtual services via linked groups of LiDAR sensors, a laser-based remote sensing procedure that function as a feeder network. This section presents a dynamic concept for high-quality reconstructions and optimal distribution of VS Pipeline (VSPs) using a preference-driven game-theoretic strategy and coexisting differential game [55].

In the field of drug discovery, such modeling balances efforts to predict and augment protein ligand interactions (PLIs). These PLIs are crucial for drug discovery, but expensive, time-consuming in vitro experiments impose computational prediction techniques, particularly machine learning-based [56].

Clark et al. [57] reports that *c*omplex structural determination techniques like MicroED further sustain precise PLI analysis. MicroED, a technique involving electron cryo-microscopy, has several applications in the field of drug discovery. It assists in evaluating molecular structures, facilitating rational drug design and development. Ultimately, Game-theoretic approaches develop protein–ligand interaction by integrating computational modeling with complex structural system to sustain competent drug design.

3.3 Deep learning (DL)

Techniques in artificial intelligence have reported efficacy in various applications, such as drug discovery in the pharmaceutical field. DL techniques are highly prevalent in drug design and generative models, for engineering bio-molecules. The deep learning landscape encompasses several powerful architectures beyond traditional neural networks. Graph Neural Networks (GNNs) operate directly on graph structures, making them ideal for molecules represented as graphs (atoms as nodes, bonds as edges). They learn by passing and transforming information between connected nodes, capturing the topological relationships crucial for predicting molecular properties. Transformers utilize self-attention mechanisms to weigh the importance of different parts of sequential input data. While foundational in NLP for text, they are equally powerful for biological sequences (proteins, DNA) and SMILES strings, enabling context-aware modeling of long-range dependencies. Diffusion models, inspired by non-equilibrium thermodynamics, learn to generate data by progressively adding noise to training data and then learning to reverse this process. In drug discovery, they are used for high-fidelity *de novo* molecular design and protein structure generation (e.g., RFdiffusion).

Additionally, DL techniques have significant achievement in AI research, particularly in natural language processing, image, and voice recognition. Its application in pharmaceutical research has outgrown bioactivity predictions, dealing various drug discovery limitations, such as bioactivity prediction, molecular design, synthesis forecasting, and biological image analysis [37, 38]. Advancing on these foundational applications, the possibilities of DL in pharmaceutical research has grown significantly in recent years.

DL techniques, including Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Variational Autoencoders (VAEs), and Generative Adversarial Networks (GANs), have found applications in several areas of drug design. Instances include RNN-driven variation autoencoders for molecular design, natural language tasks, bioactivity forecasting, and image-centric profiling of bioassays utilizing CNNs [39].

This growing multi-functionality in DL applications is closely associated with broader advances in computational biology and data science. ML and Artificial intelligence are enhancing the utilization of targets in various domains, reducing time, labor, and resources, and enabling the identification of therapeutically useful molecules [40]. Consequently, DL continues to widen its function in pharmaceutical research, offering advanced tools that enhance drug discovery and enable new solutions to challenges.

3.3.1 Neural networks

Deep learning leverages multi-layered neural networks to automatically learn hierarchical representations from complex data. Several core architectures form the foundation for most modern AI applications in drug discovery, each suited to different data types and tasks.

Artificial neural networks (ANNs), also known as multi-layer perceptrons (MLPs), serve as the fundamental building block. They consist of interconnected layers of nodes that transform input data (e.g., molecular fingerprints) into predictions for tasks like bioactivity classification.

Convolutional neural networks (CNNs) excel at processing data with spatial or grid-like topology. In drug discovery, they are not only used for analyzing cellular images in phenotypic screening but are also adept at processing 3D molecular structures and protein binding pockets by applying learnable filters that detect local patterns and features [41].

Recurrent neural networks (RNNs), including Long Short-Term Memory (LSTM) networks, are designed for sequential data. Their internal memory allows them to process information where context and order are critical, making them invaluable for analyzing biological sequences (e.g., protein or DNA sequences) and SMILES strings for molecular property prediction [38].

Graph neural networks (GNNs) operate directly on graph-structured data, making them uniquely suited for molecules represented as graphs (atoms as nodes, bonds as edges). They learn by passing and transforming information between connected nodes, enabling highly accurate predictions of molecular properties and drug-target interactions by capturing complex topological information [42].

Transformers, originally developed for natural language processing, utilize a self-attention mechanism to weigh the importance of different parts of the input data. They have been powerfully adapted for biological sequences (e.g., protein language models like ProtBERT) and molecular representations, demonstrating state-of-the-art performance in tasks ranging from protein function prediction to molecular property forecasting [38, 42].

These core architectures provide the discriminative power for a wide range of predictive tasks in drug discovery, from virtual screening and ADMET prediction to protein function annotation.

3.3.2 Generative deep learning

Deep learning architectures can be broadly categorized into discriminative models, which learn to classify or predict from data, and generative models, which learn the underlying distribution of data to create new examples. The following sections explore both categories and their applications in drug discovery: quantitative structure-activity relationship (QSAR) modeling and image-based phenotypic screening. In QSAR modeling, DL architectures particularly Graph Neural Networks surpass traditional machine learning by automatically learning relevant features from raw molecular structures, capturing complex, non-linear relationships that improve the prediction of bioactivity and physicochemical properties [43]. This enables more accurate virtual screening of large compound libraries. Simultaneously, in phenotypic drug discovery (PDD), convolutional neural networks (CNNs) have revolutionized the analysis of high-content cellular images. These models can identify subtle, drug-induced morphological changes that serve as powerful readouts of therapeutic potential [44, 45]. For example, in parasitology, DL models have been applied to classify pathogen phenotypes from video recordings based on motion and morphological features, enabling automated quantification for drug screening [46]. While phenotypic screening presents challenges in target deconvolution, the integration of AI-driven image analysis with traditional QSAR models provides a comprehensive strategy for prioritizing compounds with desired biological effects [47].

3.4 Natural language processing (NLP)

Natural Language Processing (NLP) has become an essential tool in drug development, enabling the extraction of meaningful information from unstructured text. It supports a wide range of applications, including biomarker identification, patient-trial matching, and literature-based hypothesis generation. Model-Informed Drug Development (MIDD) increasingly utilizes transformer-based models, supported by libraries (e.g., Hugging Face, sparkNLP) and open-source platforms (e.g., DisGeNet, Transmol), to enhance decision-making [58].

The application of NLP in pharmaceutical research is demonstrated by several studies. Nagalakshmi et al. [59] used ML and NLP on a large online pharmacy dataset, achieving 97% accuracy in medical treatment forecasting. ChemBERTa, a domain-adapted, pretrained NLP model, has been effectively applied in drug discovery, with architectures like DNN showing superior predictive ability using AUROC metrics [60]. Furthermore, Harrison & Sidey-Gibbons [61] employed NLP to analyze patient responses to medications like Levothyroxine and Apixaban, using unsupervised ML for drug comparability and supervised ML to predict patient ratings.

3.4.1 Literature mining, chemical entity recognition

A critical first step in biomedical NLP is Chemical Entity Recognition, which automatically identifies and classifies chemical compounds and drug names in text. The NLM-Chem corpus, for instance, is a valuable resource containing 150 full-text articles annotated with approximately 5,000 chemical entities, facilitating the development of automated annotation systems for PubMed and PMC [62, 64]. Such curated corpora are foundational for training advanced ML models. Techniques like Knowledge Graphs (KGs), entity recognition, and relationship extraction are increasingly employed to streamline the costly drug discovery process by structuring knowledge from millions of publications [63]. Continued advancements in hybrid and ML-based methods, such as the SpaCy ML model mentioned by Ramachandran & Arutchelvan [65], are enhancing the accuracy of named entity detection in medical texts.

3.4.2 BioBERT and SciBERT in pharmacology

Transformer models like BERT (Bidirectional Encoder Representations from Transformers) have revolutionized NLP tasks. In pharmacology, BERT-based models are used to discover drug mechanisms and interactions from large-scale text corpora, aiding in drug repurposing and the evaluation of biological mechanisms [66]. Domain-specific variants such as BioBERT (trained on biomedical literature) and SciBERT (trained on scientific corpus) have demonstrated superior performance in scientific literature classification and knowledge extraction, achieving higher F1-scores in tasks like relation extraction [67, 68]. For example, ValizadehAslani et al. [69] used a Relation BioBERT model with a Bidirectional LSTM to accurately identify and categorize drug-drug interactions (DDIs), thereby improving patient safety. Similarly, KafiKang & Hendawi [70] showed that an R-BioBERT and BLSTM framework could classify DDIs with high precision and superior F-scores.

3.4.3 Challenges and limitations

Despite its promise, the application of NLP in drug discovery faces significant challenges. Data scarcity is a primary hurdle, as high-quality, labeled datasets for specialized biomedical sub-domains are limited and difficult to create. This leads to the second challenge: high annotation costs, which require significant investment from domain experts (e.g., biologists, pharmacologists) to manually curate and validate textual data. Finally, model interpretability remains a critical limitation; while complex transformer models like BioBERT achieve high accuracy, their "black-box" nature makes it difficult to understand the reasoning behind specific predictions, raising concerns for clinical application and regulatory approval. Addressing these limitations through the development of richer biomedical corpora, semi-supervised learning techniques, and explainable AI (XAI) frameworks is essential for the broader and more reliable adoption of NLP in the pharmaceutical industry.

4 Data sources and tools

4.1 Public databases

In the field of chemistry, databases are among the most significant instruments for integrating information regarding molecular structure, physicochemical properties, bioactivity, synthesis reactions, and toxicology. In the field of drug discovery, the existence of structured and interconnected databases facilitates researchers' ability to efficiently identify candidate molecules, predict pharmacokinetic properties and toxicity, design efficient synthesis pathways, and evaluate their potential biological interactions. These databases can then be classified based on their information and utility to facilitate mapping the role of each resource in supporting the overall drug discovery process [71].

4.1.1 Database for general chemical structure and properties

The database in this category serves as an initial point of reference for searching chemical information containing compounds from various classes that can be widely used in various fields of chemistry. The database contains a variety of information about each compound, including molecular structure in 2D and 3D formats, physicochemical parameters such as melting point, boiling point, pKa, logP, solubility, official identification numbers such as CAS, InChI, and SMILES, as well as links to external sources and supplier information. In certain instances, databases have been observed to incorporate biological or toxicological data, thereby facilitating the acquisition of exhaustive information regarding the specific compound under consideration [72].

PubChem is one of the most comprehensive and frequently used databases. It collects data on more than 322 million chemical compounds, along with their physicochemical properties, suppliers, and bioactivities [73]. This data is obtained from more than 748 sources, including scientific literature, suppliers, and other databases [72]. A similar database is ChemSpider, which integrates data from over 275 sources and offers search features based on structure, substructure, or molecular similarity [74]. Similar databases include the German chemical database RÖMPP and the Japanese chemical database J-GLOBAL Nikaji. There are also databases that provide information about chemical compound suppliers such as Chemical Book and MolPort. This database present chemical compound data as commercial catalogs that include prices, suppliers, and technical specifications of the compounds. ChemExper is another database that provides information about chemical compound suppliers [75]. It presents chemical data based on CAS numbers, structure, and supplier availability. Furthermore, there is ZInc, which is a resource that focuses on providing information about commercially available molecules for virtual screening [76]. In this category are databases that provide concise data for verifying compounds, such as ChemIndex and Common Chemistry. These databases focus on compounds commonly used in education and industry. Another database is Unichem functions as an intermediary between databases by identifying the molecular identity of 41 different data sources, thereby facilitating cross-referencing [77].

This category also encompasses numerous databases that are more specialized yet still provide extensive coverage. One notable example is ACToR (Aggregated Computational Toxicology Resource), which presents chemical data from the US EPA alongside toxicology, exposure, and regulatory information [78]. Another database, DDB (Dortmund Data Bank), focuses on presenting data on the physical properties of pure compounds, mixtures, and hydrate gases relevant to chemical process engineering [79]. HugeMDB offers a collection of small-molecule data accompanied by three-dimensional conformer models, which can be utilized for computational purposes. eMolecules and Mcule are capable of conducting molecular searches for virtual screening, a process that is informed by considerations of commercial availability [80].

The databases in this category facilitate researchers' efforts in conducting preliminary molecular analysis, identifying candidates for synthesis or acquisition, verifying structures, calculating predictive parameters, and even developing experimental or modeling strategies based on the completeness of the available data [80]. A concise summary of key public databases, categorized by their primary use, is provided in Table 3.

4.1.2 Database for organic compounds & natural products

In this category, the database focuses on providing information on organic compounds derived from primary and secondary metabolites, microbial natural products, phytochemicals from plants, and other organic compounds produced by biological systems. While the information presented typically includes molecular structure data, stereochemistry, biological or pharmacological activity, biological sources (species, organ, or tissue where the compound is found), as well as information on biosynthetic or degradation pathways, the focus of this category is on organic compounds [81].

One notable database in this category is HMDB, which provides information on human metabolites along with biochemical information, clinical data, and NMR and

Table 3 Summary of public databases for drug discovery

Category	Database name	Primary use	Website URL
General Chemical	PubChem	Comprehensive repository of chemical compounds, substances, and bioactivities	https://pubchem.ncbi.nlm.nih.gov
	ChemSpider	Community resource of chemical structures & properties from > 275 sources	http://www.chemspider.com
	ZINC	Commercially available compounds for virtual screening	https://zinc.docking.org
Natural Products	HMDB (Human Metabolome Database)	Detailed information on human metabolites	https://hmdb.ca
	Super Natural II	Database of natural compounds and their suppliers	http://bioinf-applied.charite.de/supernatural_new
	NPAtlas	Database of microbial natural products	https://www.npatlas.org
Crystallography	Cambridge Structural Database (CSD)	Repository of organic and organometallic crystal structures	https://www.ccdc.cam.ac.uk
	Crystallography Open Database (COD)	Open-access collection of crystal structures	http://www.crystallography.net/cod
	Protein Data Bank (PDB)	3D structures of proteins, nucleic acids, and complexes	https://www.rcsb.org
Pharmacology & Targets	DrugBank	Detailed drug data with drug-target information	https://go.drugbank.com
	ChEMBL	Bioactivity data of drug-like molecules	https://www.ebi.ac.uk/chembl
	BindingDB	Binding affinities of drug-target interactions	https://www.bindingdb.org
Toxicology & Regulation	CompTox Chemicals Dashboard (EPA)	Chemical data with toxicology and risk assessment	https://comptox.epa.gov/dashboard
	T3DB (Toxin Target Database)	Information on toxins and their biological targets	http://www.t3db.ca
	CEBS (Chemical Effects in Biological Systems)	Toxicogenomics database from the NIEHS/NIH	https://cebs.niehs.nih.gov
Biology & Biochemistry	UniProt	Comprehensive resource for protein sequence and functional data	https://www.uniprot.org
	KEGG (Kyoto Encyclopedia of Genes and Genomes)	Database of biological pathways, genomes, and diseases	https://www.genome.jp/kegg
	BRENDA	Comprehensive enzyme information and kinetic data	https://www.brenda-enzymes.org

MS spectra. In addition to databases focused on human metabolites, there are databases that provide data from other organisms [82]. Another database is BMDB, which provides information on metabolites from cattle, and YMDB, which provides information on metabolites from yeast [231, 84]. A further database that furnishes information on metabolites is GMD, which provides GC/MS data on a metabolite [85]. Other databases in this category include KNApSAcK. KNApSAcK is a valuable resource, as it provides data on natural compounds along with their producing organisms and chemical identities [86]. A similar database is Super Natural II, which furnishes data on the most substantial natural compounds, in addition to their respective suppliers [87]. Npatlas is a database that also provides data on natural products, albeit exclusively those derived from microorganisms and fungi [88].

Furthermore, there are databases that focus exclusively on certain organic compounds. One such example is the Carotenoids Database, which provides information on the structure, properties, and organisms that produce carotenoid compounds [89]. Additionally, BIAdb offers data on benzylisoquinoline alkaloids, their sources, and their activities [90]. For phenolic compounds, there is Phenol-Explorer, which provides data on polyphenols derived from food [91]. Additionally, there are databases that focus on presenting data from ligand-protein interactions, such as BindingDB and BindingMOAD.

These databases have applications in a variety of fields, including clinical metabolic research, which functions as a disease biomarker, chemical biodiversity research, which is useful for various fields such as pharmacy and nutrition, the optimization of the production of secondary metabolites sourced from organisms, and the discovery of drugs from natural materials [92].

4.1.3 Database for crystallography & solid-state structure

The focus of databases in this category is the provision of crystal structure data for solid compounds. The information contained in these databases includes, but is not limited to, lattice parameters, crystal space groups, atomic coordinates in unit cells, polymorphism, and other structural properties [92].

Notable databases within this category include the Cambridge Structural Database (CSD), which offers a repository of organic and organometallic molecules, accompanied by comprehensive information on their crystal structures [93]. COD (Crystallography Open Database) offers a repository of crystal structure data for both organic and inorganic compounds [94]. A similar database, ICSD (Inorganic Crystal Structure Database), is exclusively dedicated to the provision of crystal structure data for inorganic compounds [95]. A further database that furnishes data on inorganic compounds is Atom Work, which encompasses data for industrial applications and materials engineering [96]. A database, known as Crystal Works, has been developed to facilitate cross-referencing by combining data from CSD, ICSD, and COD.

The utilization of these databases is instrumental in the design of functional materials, including semiconductors, catalysts, and batteries. It is also employed in the control of drug polymorphism for the purpose of optimizing solubility and stability. Furthermore, these databases are utilized in the modeling of the mechanical and electronic properties of materials. Finally, they play a crucial role in the research of structure-property relationships, which is essential for innovation in the field of materials chemistry [97].

4.1.4 Database for spectroscopy and physical properties

The database in this category contains information from experimental measurements and compound characterization, including spectra from various analytical techniques (NMR, IR, Raman, MS, and UV-Vis) as well as physical and thermodynamic property data (solubility, pKa, viscosity, Henry's constant, and thermochemical data) [98].

In the field of chemical spectra, there are several databases that provide such information. One such database is SBDS, which presents IR, Raman, MS, and NMR spectrum data for organic compounds [99]. This data can be used for pattern matching. The NIST Webbook is another database that provides extensive information. It offers data on spectra, thermochemical data, physical constants, and reaction information. This database is frequently used in both research and industry [100]. Another database, MoNA (Mass Bank of North America), exclusively provides mass spectrum data for organic compounds, and it relies on community contributions to obtain data for its database. Furthermore, there are databases that focus exclusively on providing specific spectral results. One example is NMRShiftDB, which provides information about ^{1}H and ^{13}C NMR spectra. These spectra are useful for the analysis of organic compounds [101]. Another database that displays NMR data is BMRB, which provides NMR spectra of biomolecules [102]. Another database is UV/VIS Spectral Atlas, a database that exclusively provides atmospheric gas absorption spectrum data [103].

Meanwhile, database with a focus on physical data, DETHERM is recognized as one of the most comprehensive databases available. This database supplies thermophysical property data for compounds and mixtures [104]. A database similar to DETHERM is ILThermo, which furnishes physical property data for ionic liquids and their mixtures [105]. Furthermore, there are databases that focus exclusively on providing a specific physical property of compounds, such as IUPAC Digitized pKa Dataset, evaluated kinetic data, Henry's law constants, and the IUPAC-NIST Solubility Database. IUPAC digitized pKa dataset is a database that provides information on digital datasets of dissociation constants (pKa). Evaluated kinetic data is a database that provides data on reaction rate constants [106]. Concurrently, Henry's Law Constants offers data derived from multiple literature sources [107]. The IUPAC-NIST Solubility Database is a comprehensive repository of solubility data for various substances [108].

The database is primarily utilized for the identification of compounds through spectrum matching, the development of quantitative analysis methods, the validation of the quality of raw materials in the pharmaceutical and chemical industries, the modeling of physical properties for process optimization, and the validation of research experiment results [109].

4.1.5 Databases for pharmacy, medicines & biological targets

The databases in this category provide information on bioactive molecules, approved drugs, drug candidates in development, and their biological targets. These databases offer data on molecular structure, bioactivity, pharmacokinetics, pharmacodynamics, structure-activity relationships (SAR), and interactions with their target proteins [110].

In the context of drug data, several databases have been developed for the purpose of providing such information, including DrugBank, DrugCentral, Merck Index, and RxNav. DrugBank is a database that provides comprehensive data by combining chemical, pharmacological, mechanism of action, and regulatory status information for thousands of drugs [111]. Meanwhile, Merck Index is an encyclopedic database on drugs and drug compounds. Concurrently, DrugCentral and RxNav are databases that facilitate the identification of pharmaceutical agents, generic equivalents, and pharmacological interactions [112, 113].

In the database category that also contains biological target information, Guide to Pharmacology is a database that offers curated annotations of drug compounds and their biological targets, while SuperDRUG2 offers more extensive information with additional data on dosage and side effects [114, 115]. Meanwhile, TDR Targets is exclusively focus on data tropical disease drugs and their respective targets [116]. CHEMBL is a comprehensive database that provides information on the bioactivity of drug-like molecules against defined biological targets [117]. Within this category, there are also databases that exclusively focus on providing biological target data. For examples such as PHAROS, TTD (Therapeutic Target Database), and Ki Database, they present information on therapeutic targets along with their active compounds or related ligands.

Additionally, there are databases that provide specific information relevant to the research and development of pharmaceuticals. SureCHEMBL is a comprehensive database that offers a wealth of information on compounds derived from pharmaceutical-related patent documents. This unique resource facilitates searches for the most recent advancements in the field of medicine, providing researchers and practitioners with valuable insights of therapeutic agents and their potential applications [118]. DTP/NCI is a valuable resource, providing comprehensive information on drug candidates utilized in the treatment of cancer. LINCS (Library of Integrated Network-Based Cellular Signatures) furnishes data on cellular responses to thousands of bioactive molecules [119]. Additionally, a database has been developed that provides information on molecules that can be used in clinical trials, namely Clival. The availability of these various databases serves as an important resource supporting researchers in the process of drug discovery, development, and optimization, as well as in understanding the interactions between bioactive molecules and biological targets in greater depth [120].

4.1.6 Database for toxicology & regulation

The databases in this category contain comprehensive information on chemical safety, health risk assessment, and compliance with international regulations. The information presented includes hazard classification, exposure limits, toxic properties, legal status in various jurisdictions, and safety guidelines [121].

In the context of chemical safety information, certain databases, including the Chemical Carcinogenesis Research Information System (CCRIS), offer data concerning carcinogenic compounds. Concurrently, the RTECS (Registry of Toxic Effects of Chemical Substances) offers data regarding the toxic effects of chemicals [122]. A database that provides similar information is CompTox, developed by the United States Environmental Protection Agency (EPA). CompTox integrates data on physical properties, toxicology, and hazard prediction models for chemicals [123]. A further database, the International Chemical Safety Cards (ICSC), offers information regarding the hazards and safety of chemicals for workers [124]. The NIOSH Pocket Guide is a compendium of information regarding chemical exposure limits in the workplace and safety guidelines [125]. Furthermore, another database is T3DB (Toxin and Toxin Target Database), has been developed to provide information on toxins and their corresponding biological targets [126].

Within the regulatory domain of chemicals, several databases are available, including ECHA/REACH, which provides data on chemicals registered in Europe [127]. A similar database, CosIng is a valuable resource that provides information regarding cosmetic ingredients and their regulatory status in Europe [128]. The ENCS and JECDB are databases that provide information on chemicals that have been registered in Japan. The FDA SRS (Substance Registration System) is a comprehensive database that houses information regarding chemicals that are subject to regulation by the FDA. The Global Substance Registration System (G-SRS) is a comprehensive database that houses international chemical registration numbers [129]. Additionally, a comprehensive database, known as LOLI (List of Lists), has been developed to systematically compile and consolidate regulatory frameworks from multiple nations.

The database has a number of applications, including industrial risk analysis, the preparation of Safety Data Sheets (SDSs), environmental monitoring, chemical trade legal compliance, and the determination of occupational safety policy [121].

4.1.7 Database for biology and biochemistry

Databases in this category facilitate the integration of chemical data with biological contexts, including enzymes, proteins, metabolic pathways, and biomolecular interactions. Databases in this category play an instrumental role in biotechnology, pharmacology, and biological systems research [130].

With reference to the subject of enzymes and proteins, there are several databases that provide different types of data. For example, BRENDA is the largest enzyme database, covering kinetic data, substrates, inhibitors, and reaction conditions [131]. Uniprot is the primary source of protein data with functional annotations [132]. The Protein Data Bank (PDB) is a comprehensive repository of structural data on proteins and their complexes [133]. Phosida is a database that focuses on protein phosphorylation sites. STITCH is a comprehensive database that offers insights into protein-protein interactions [134]. Another databases are available for lipid-related information, including LipidBank and LMSD, which focuses on lipid structures, and SwissLipids, which includes information on lipid metabolism [135].

With regard to information on metabolic pathways, one example is KEGG, which maps metabolic pathways, genes, proteins, and related compounds [136]. MetaCyc is a comprehensive database that houses information regarding metabolic pathways and their associated metabolites [137]. Another database, METLIN furnishes mass spectrometry data concerning metabolites [138]. Additionally, a database, known as the Serum Metabolome DB, has been developed to provide comprehensive data on metabolites present in blood. For databases containing information on metabolic pathway models, there are MetaNetX and ModelSeed. Meanwhile, the biochemical reactions occurring within metabolic pathways, Rhea is a relevant database [139].

The database is predominantly utilized for metabolic engineering, biological pathway analysis to comprehend diseases or drug responses, enzyme design for biocatalysis, and the development of biological function prediction models [140].

4.1.8 Database for chemical reaction

The focus of the databases in this category is on data concerning chemical reactions, synthesis routes, and reaction kinetics. The information provided in this category encompasses reaction schemes, experimental conditions, yields, and reference literature [141].

In regard to information on chemical reactions, one of the most extensive databases is SciFinder, which links compounds, reactions, and scientific publications. Similar databases, including Reaxys, SaguaroChem, and SPRESIweb, provide curated reaction data derived from both literature and patents, encompassing condition optimization. Another database, ORD (Open Reaction Database) is an open repository for reaction data that can be downloaded and processed computationally [141]. In addition to data on chemical reactions, several databases also provide data on synthesis routes. For example, Organic Syntheses (OrgSyn) provides verified organic synthesis procedures, and Chematica (Synthia) uses artificial intelligence to plan automatic synthesis routes [142, 143]. The database is widely used for synthesis planning, production route optimization, discovery of alternative synthesis routes, and organic chemistry learning with real reaction examples. A summary of key public databases, their primary uses, and access links is provided in Table 4 for quick reference.

4.1.9 Integration of curated and high-throughput screening data

The drug discovery process strategically leverages both high-throughput screening (HTS) and curated data to balance scale with precision. HTS, enabled by cutting-edge robotics and automation, allows for the rapid testing of massive compound libraries, generating vast datasets that provide a broad survey of chemical space [144, 145]. However, this approach is often susceptible to noise, including false positives/negatives due to compound interference or assay artifacts, and requires sophisticated data management and analytical tools [146].

Curated data, in contrast, are derived from a rigorous process of refinement and validation, resulting in high-quality, context-rich datasets with minimized errors [147]. This quality makes curated data indispensable for building reliable AI models, enabling robust cross-data analyses like pharmacological networking and sophisticated QSAR

Table 4 AI-Driven drug discovery platforms comparison

S.*N*	Platform	Company	Technology	Features	Achievements	Success rate	Timeline	Cost reduction	References
1.	**Exscientia**	Exscientia (UK)	Deep Learning + Evolutionary Algorithms	Centaur Chemist/Biologist systems	First AI-designed drug in trials (12 months)	High	12 months to trials	Up to 40%	[8, 38]
2.	**Atomwise (AtomNet)**	Atomwise (US)	Deep CNNs	3 + trillion compound library screening	70%+ success rate in hit identification	70%+	Rapid screening	Significant	[40, 42]
3.	**Insilico Medicine**	Insilico Medicine	GANs + Machine Learning	Generative molecular design	AI-designed cancer drug trials	Moderate	12–18 months	Up to 40%	[41, 45]
4.	**Schrödinger LiveDesign**	Schrödinger	Molecular Modeling + AI	High accuracy modeling	High computational accuracy	High	Variable	Moderate	[27, 46]
5.	**Cyclica**	Cyclica	Molecular AI	Polypharmacology focus	Multi-target approaches	Moderate	Variable	Moderate	[48, 55]

studies that reveal relationships between chemical scaffolds and disease pathways [147, 148].

4.2 AI platforms and frameworks

The rapid development of AI has given rise to platforms and frameworks specifically designed to handle complex data in chemistry and molecular biology. AI platforms provide the necessary computing infrastructure and algorithm libraries, as well as database integration. Frameworks, on the other hand, offer a foundation that enables the flexible development, training, and deployment of machine learning or deep learning models. Adapting this technology for chemical research involves optimizing data representations, such as SMILES, molecular graphs, and protein sequences, and developing predictive models for molecular properties, functions, and interactions. AI platforms and frameworks can be classified by primary functionality, input and output types, enabling AI applications to be grouped into four categories: Molecular Property Prediction; Molecular Representation and Generation; Protein Structure and Function Prediction; and Cheminformatics Toolkits with AI [149150]. A comparison of prominent AI-driven drug discovery platforms, highlighting their technologies and achievements, is provided in Table 4.

4.2.1 Molecular property prediction

The AI platforms and networks in this category are designed to predict the physicochemical properties of molecules or compounds, such as solubility, toxicity, and bioactivity, based on their chemical structure. These platforms and networks process data such as SMILES strings, which are graphical representations showing atomic bonds, and chemical descriptor data. The system then generates numerical values or labels describing the molecule's or compound's chemical properties or bioactivity [151].

One example of an AI platform and network in this category is Deepchem. DeepChem is an open-source machine learning platform based on Python. This platform offers modules for predicting quantitative structure activity relationships (QSAR), predicting protein–ligand interactions, and estimating material properties. In addition to DeepChem, ChemProp employs graph neural networks (GNNs). This system has been developed for the purpose of extracting topological information from a molecule's SMILES structure and subsequently predicting its physicochemical properties, including solubility, potential, and bioactivity [152]. These platforms have gained significant popularity among researchers in fields such as chemistry, biology, and pharmaceutical sciences, owing to their ability to predict the properties of molecules under development.

4.2.2 Molecular representation & generation

AI platforms and networks in this category are designed to present informative molecules (embeddings) and can be used for de novo drug design. Platforms operating within this category utilise a "chemical language" approach, whereby molecules are analysed within the context of the platform's understanding of the subject matter. This adaptation of natural language processing (NLP) techniques enables the platform to comprehend and respond to contextual information [153].

MolBert is a platform that implements the transformer-based BERT model to learn millions of SMILES molecules, which are then used to create molecular embeddings.

These embeddings are then used for classification, compound search, or prediction of physicochemical properties [154]. Another platform is MegaMolBART, developed by NVIDIA and AstraZeneca, which adapts transformer encoder-decoder techniques to design new molecules with desired properties (de novo drug design) [155]. Additionally, NVIDIA has developed another system known as BioNeMo. These platforms are utilised for foundation models in the domains of chemistry and biology [156]. It is evident that the platforms have the capacity to represent molecules, predict their properties, and design molecules, based on the foundation models.

4.2.3 Protein structure & function prediction

AI platforms and networks in this category are focused on predicting the three-dimensional structure, function, or interaction of proteins. The data processed in such platforms consists of amino acid sequences, which can be supplemented with partial structural data or related experimental information to strengthen the data [157].

AlphaFold is a prominent example of a platform in this category that is frequently employed. This platform adapts DeepMind's deep learning model, which is utilised to predict the 3D structure of proteins from amino acid sequences, achieving a level of accuracy that approaches that of X-ray crystallography experiments [158]. Prot Trans is a platform that is utilised for the prediction of protein function, classification, and interactions. This platform employs a protein language model approach that has been trained on a substantial number of protein sequences, estimated to be in the billions [159]. A similiar platform, Atom Net, employs a convolutional neural network (CNN) model to predict ligand-protein interactions based on three-dimensional structures [6]. This platform is extensively utilised in the domain of new drug discovery through structure-based drug design methodologies.

4.2.4 Cheminformatics toolkits with AI

AI platforms and networks in this category include multifunctional computational chemistry platforms equipped with AI modules to expand their functionality. Platforms in this category function as integrated platforms, capable of handling various types of chemical data, performing format conversions, and integrating machine learning models for advanced analysis [160].

OpenBabel, when combined with AI plugins, is one of the platforms or toolkits utilised for the conversion of molecular formats, with a capacity to support over 100 file formats. Integration with AI plugins can be used for predicting molecular properties or optimising molecules [161]. Another platform is Schrödinger's DeepChemistry, a suite of computational chemistry applications with AI modules to accelerate molecular design, drug optimisation, and molecular dynamics simulations [162]. These platforms are extensively utilised in a commercial context within the pharmaceutical industry.

4.3 Data curation and preprocessing

The performance of AI models in drug discovery is fundamentally constrained by the quality of the underlying data. Effective data curation and preprocessing are therefore critical steps to mitigate common issues that can introduce noise and bias. Key challenges include missing values in experimental readouts, batch effects from technical variation across experiments, inconsistent annotations and identifiers (e.g., different naming conventions for the same compound or target), and data noise from experimental error or compound interference [146, 220].

To address these issues, a multi-faceted strategy is employed. Standardization protocols are applied to normalize compound structures (e.g., using InChI keys), correct stereochemistry, and standardize biological target nomenclature across datasets. For handling missing data, imputation methods ranging from simple mean/median substitution to more sophisticated k-nearest neighbors or model-based imputation are used, though their application must be carefully validated to avoid introducing artifactual patterns. Normalization and batch effect correction techniques, such as ComBat or Z-score transformation, are essential when integrating data from multiple sources to ensure that biological signals are not confounded by technical variation [222].

The curation workflow itself often involves a hybrid of automated and manual processes. Automated pipelines can flag outliers, check for structural integrity, and map entities to standardized databases. However, expert manual curation remains indispensable for resolving complex ambiguities, validating critical data points, and adding rich contextual annotations that automated systems might miss. This rigorous, iterative process of cleaning, standardizing, and enriching datasets is a prerequisite for building reliable, generalizable AI models that can truly accelerate drug discovery.

5 Applications in drug discovery pipeline

5.1 Target identification and validation

5.1.1 Target identifications

Analysis of complex biological networks and disease mechanisms. Machine learning algorithms can process vast datasets from genomics, proteomics, and other omics technologies to identify novel therapeutic targets that might otherwise remain undiscovered [163]. Network-based approaches utilize AI to analyze protein-protein interaction networks, metabolic pathways, and gene regulatory networks to identify key nodes that represent potential drug targets [164]. These methods can reveal non-obvious relationships between biological entities and disease states, leading to the discovery of innovative therapeutic approaches. The convergence of AI with other disruptive technologies is creating powerful new paradigms for drug discovery. These integrations enhance the precision, speed, and success rate of the entire pipeline, as illustrated in Fig. 3.

5.1.2 Hit identification and virtual screening

Virtual screening has been revolutionized by deep learning architectures that go beyond traditional machine learning. Graph Neural Networks (GNNs), which operate directly on the molecular graph structure, have become a powerful tool for QSAR modeling. They automatically learn relevant features from atoms and bonds, leading to superior prediction of bioactivity compared to methods relying on pre-defined molecular descriptors [165]. For example, the ChemProp model has demonstrated state-of-the-art accuracy in predicting molecular properties and screening for active compounds [166].

Similarly, Convolutional Neural Networks (CNNs) can be applied to 3D structural data of protein targets to perform structure-based virtual screening. Models like AtomNet predict binding affinity by analyzing the spatial arrangement of atoms in a binding pocket, significantly accelerating the identification of potential hits [165].

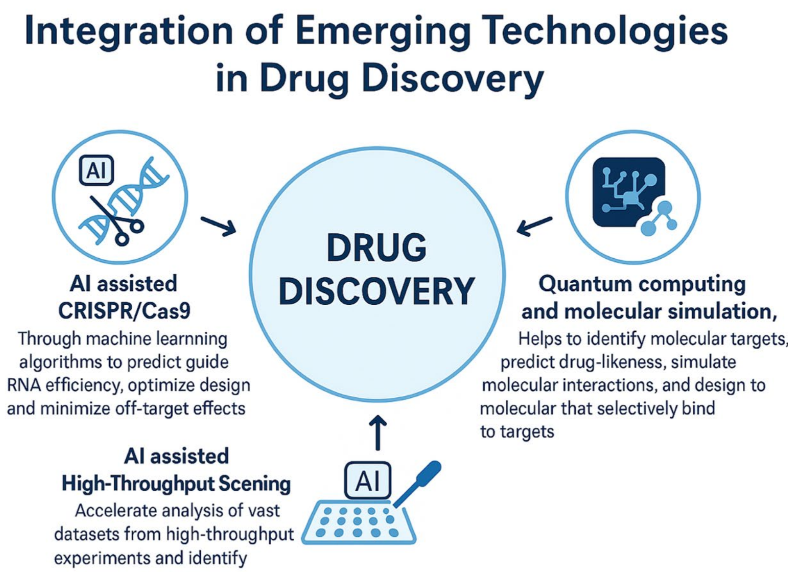

Fig. 3 Integration of emerging technologies in drug discovery

5.1.3 Pharmacophore modeling

Pharmacophore modeling enhanced by AI enables the identification of essential molecular features required for biological activity, facilitating the design of new compounds with improved properties [167]. Machine learning algorithms can automatically derive pharmacophore models from experimental data, reducing the time and expertise required for model.

5.1.4 Lead optimization

AI technologies have transformed lead optimization processes by enabling more accurate prediction of molecular properties and biological activities [168]. Machine learning models can predict absorption, distribution, metabolism, excretion, and toxicity (ADMET) properties, allowing researchers to optimize compounds before synthesis and testing [169].

5.1.5 QSAR modeling

QSAR modeling enhanced by machine learning techniques provides improved structure-activity relationships that guide medicinal chemistry efforts [170]. Deep learning approaches can capture complex, non-linear relationships between mole cular structure and biological activity that are difficult to identify using traditional methods [171].

5.1.6 ADMET prediction

The accurate prediction of ADMET properties is crucial for reducing late-stage attrition. Deep learning models have shown remarkable success in this domain by modeling complex, non-linear relationships. Multitask Deep Neural Networks can simultaneously predict multiple ADMET endpoints (e.g., solubility, permeability, toxicity) from a single molecular structure, leveraging shared representations to improve overall accuracy and efficiency [172–174]. Furthermore, recurrent neural networks (RNNs) like LSTMs can model metabolic pathways as sequences of transformations, offering a dynamic approach to predicting drug metabolism and potential drug-drug interactions.

5.1.7 Clinical trial optimization

AI applications in clinical trial design and management offer significant potential for improving the efficiency and success rates of clinical development. Machine learning algorithms can identify optimal patient populations for specific therapies, predict treatment responses, and design adaptive trial protocols that adjust based on interim results [175]. Patient stratification using AI-driven biomarker analysis enables the identification of subpopulations most likely to benefit from specific treatments, supporting the development of personalized medicine approaches [176]. These capabilities are particularly valuable in oncology, where patient heterogeneity significantly impacts treatment outcomes.

5.2 Protein structure prediction and design

5.2.1 AlphaFold and beyond

The development of AlphaFold represents one of the most significant breakthroughs in computational biology, fundamentally changing how researchers approach protein structure prediction. AlphaFold 2's unprecedented accuracy in protein structure prediction has enabled numerous applications in drug discovery, enzyme design, and protein engineering [177].

5.2.2 AlphaFold 3

AlphaFold 3 has expanded these capabilities beyond individual proteins to predict the structure and interactions of complexes containing proteins, nucleic acids, small molecules, ions, and modified residues [178]. This advancement represents a significant step toward understanding the complete molecular machinery of cells and has immediate applications in drug design and biological research. The AlphaFold Protein Structure Database, containing over 200 million protein structure predictions, has become an invaluable resource for the global research community [179]. The database provides broad coverage of UniProt and includes specialized downloads for key organisms important in research and global health.

5.2.3 De novo protein design

RFdiffusion has emerged as a powerful tool for de novo protein design, enabling the generation of diverse functional proteins from simple molecular specifications. This diffusion-based approach has demonstrated outstanding performance across various design challenges, including unconditional protein design, protein binder design, and symmetric oligomer design [180]. The technology has shown remarkable success in experimental validation, with approximately half of tested unconditional designs expressing in a soluble form and displaying high thermostability. The accuracy of RFdiffusion predictions has been confirmed through cryogenic electron microscopy structures of designed proteins [181].

5.2.4 Protein-ligand interactions

AI technologies have significantly advanced the prediction and design of protein-ligand interactions, enabling more accurate drug design approaches. Context-aware geometric deep learning models can predict protein sequences from backbone scaffolds while considering constraints imposed by diverse molecular environments [182]. AlphaFold 3's ability to predict protein-ligand interactions with at least 50% improvement compared to existing methods represents a major advance for drug discovery applications [183]. This capability enables more accurate prediction of drug binding modes and affinities, supporting rational drug design efforts.

5.2.5 Molecular dynamics simulations

AI-enhanced molecular dynamics simulations provide detailed insights into protein dynamics, conformational changes, and drug binding mechanisms. Machine learning approaches can accelerate MD simulations and improve sampling of conformational space, enabling the study of biological processes that occur on longer timescales. The integration of AI with MD simulations supports enhanced drug design by providing detailed understanding of protein-drug interactions, binding kinetics, and resistance mechanisms [184]. These capabilities are particularly valuable for designing drugs against challenging targets such as membrane proteins and intrinsically disordered proteins [185].

5.3 Multi-omics integration and systems biology

5.3.1 Genomics and transcriptomics

The integration of genomics and transcriptomics data represents a fundamental application of AI in systems biology, enabling comprehensive understanding of gene regulation and expression patterns. Machine learning algorithms can identify regulatory elements, predict gene function, and understand the flow of genetic information from DNA to RNA [186].

5.3.2 Genome-scale models

Enhanced by AI enable the integration and interpretation of genomics and transcriptomics data, facilitating the modeling of metabolic, transcriptional, and translational reactions within organisms. These approaches support the identification of therapeutic targets and the understanding of disease mechanisms at the systems level [187].

5.3.3 Proteomics and metabolomics

AI applications in proteomics and metabolomics enable comprehensive analysis of protein expression, modifications, and metabolic profiles. Machine learning algorithms can identify biomarkers, classify disease states, and predict treatment responses based on proteomic and metabolomic signatures. The integration of proteomics and metabolomics data through AI approaches supports the development of systems-level understanding of biological processes and disease mechanisms. These capabilities are particularly valuable for precision medicine applications, where molecular signatures can guide treatment selection [188, 189].

5.4 Network biology

5.4.1 Network construction and regulatory analysis

Represent critical applications of AI in systems biology, enabling the understanding of complex interactions within biological systems. Machine learning algorithms can construct networks representing relationships between genes, proteins, and metabolites, identifying key regulatory mechanisms and pathways [190].

5.4.2 Heterogeneous multi-layered networks

(HMLNs) represent advanced approaches for integrating diverse biological data types and understanding hierarchical interactions within biological systems. These networks can infer novel biological relationships and provide insights into environmental impacts on organisms [191].

5.4.3 Personalized medicine

AI-driven multi-omics integration is enabling the development of personalized medicine approaches that consider individual genetic, molecular, and clinical profiles. Machine learning algorithms can analyze comprehensive molecular data to predict disease risks, treatment responses, and optimal therapeutic strategies for individual patients. The application of AI to personalized medicine extends beyond treatment selection to include disease prevention, early detection, and monitoring of treatment responses. These capabilities are transforming healthcare delivery by enabling more precise, individualized approaches to medical care [192].

6 Case studies and success stories

The traditional drug discovery paradigm is notoriously protracted and resource-intensive, exemplified by the multi-decade development of Taxol. Artificial intelligence is now demonstrating its potential to disrupt this timeline across multiple therapeutic modalities. This section critically examines AI's impact through case studies in small molecules, antibodies, vaccines, and drug repurposing, highlighting both its achievements and the stages of the pipeline it influences, from target discovery to candidate optimization [193].

6.1 Small molecules

Small molecules represent the most advanced category of AI-derived therapeutics, showcasing applications in *de novo* design, optimization, and novel target identification.

A) **Design and optimization**: AI accelerates the design of ligands with optimized affinity, selectivity, and pharmacokinetic profiles. For instance, Relay Therapeutics' RLY-4008, a highly selective FGFR2 inhibitor for cholangiocarcinoma, was designed using AI models informed by specific FGFR2 mutation profiles [194]. Similarly, Nimbus Therapeutics' NDI-034858, a TYK2 inhibitor for autoimmune diseases, entered Phase 3 trials following an AI-driven design that achieved superior selectivity over traditional approaches [193].

B) **Target discovery**: AI systematically mines genomic and proteomic databases to identify novel, tractable targets. Black Diamond Therapeutics engineered BDTX-1535, a precision inhibitor targeting specific EGFR mutations in lung cancer, which were first identified through AI analysis of patient genomic data [195]. Landos Biopharma's

NX-13, an NLRX1 agonist for inflammatory bowel disease, emerged from AI-driven identification of novel immunometabolism pathways [196].

C) **Integrated approaches**: Some platforms exemplify a full-stack AI approach. Insilico Medicine's INS018-055 for idiopathic pulmonary fibrosis involved AI-first target identification within fibrotic pathways, followed by generative AI to design the lead inhibitor molecule, enabling a rapid transition to clinical trials [197]. This integrated pipeline represents a significant ambition for the field.

6.2 Antibodies

AI is reducing the time and cost of antibody development by predicting structure-function relationships and identifying novel targets, moving beyond traditional iterative screening.

A) **Design and optimization**: AI models prioritize antibody sequences for enhanced affinity and specificity while minimizing immunogenicity. Biolojic Design's AU-007 is a therapeutic antibody that targets the IL-2 pathway, designed by AI to activate anti-tumor T cells without inducing deleterious cytokine release syndrome [198].

B) **Target discovery**: AI algorithms analyze complex interaction networks to uncover new immunotherapy targets. Compugen identified PVRIG as a novel target in tumors resistant to conventional immunotherapy, leading to the development of the antibody COM-701 [199]. Similarly, Hummingbird Bioscience's HMBD-001, which targets HER3/EGFR networks in cancer, was derived from an AI-driven map of tyrosine kinase receptor interactions [200].

6.3 Vaccine

n vaccinology, AI's primary impact lies in the prediction of immunogenic epitopes for both cancer neoantigens and highly mutable pathogens.

A) **Cancer vaccines**: AI models predict patient-specific neoantigens to design personalized cancer immunotherapies. Evaxion Biotech's **EVX-01** for melanoma and Gritstone Bio's **GRANITE-001** for solid tumors both utilize this approach to elicit cytotoxic T-cell responses [201, 202203].

B) **Infectious disease vaccines**: AI has been applied to identify conserved epitopes on variable viruses. While one AI-designed HIV vaccine candidate demonstrated the ability to generate novel immunogens, its eventual termination underscores the technical and immunological hurdles that remain, even with advanced computational design [193].

6.4 Drug repurposing

Drug repurposing leverages AI to find new indications for existing compounds by analyzing vast molecular, clinical, and literature datasets, capitalizing on known safety profiles to accelerate clinical translation.

A) **Neuropsychiatry**: BioXcel Therapeutics used AI to identify BXCL-501 (dexmedetomidine) for the acute treatment of agitation in Alzheimer's and schizophrenia, successfully advancing it to Phase III trials [204].

B) **Oncology: Talabostat (BXCL-701)** was repurposed as an immunomodulatory agent for prostate cancer through AI-based repositioning analysis [205].

C) **Neurology**: SOM Biotech identified SOM-3355 (bevantolol), a former beta-blocker, as a potential therapy for reducing chorea in Huntington's disease, demonstrating AI's ability to find novel mechanisms for existing chemical entities [206].

7 Emerging technologies and future directions

7.1 Quantum computing

Quantum machine learning represents a promising frontier for drug discovery applications, offering potential advantages for specific types of molecular modeling problems [207]. Quantum neural networks and variational quantum circuits have shown initial promise in applications such as molecular property prediction and protein structure analysis. Current quantum machine learning research in drug discovery focuses primarily on hit identification and novel molecule generation [208]. However, the field faces significant challenges related to the limitations of current quantum hardware, including limited qubit counts and noise issues that restrict the complexity of problems that can be addressed. The development of quantum algorithms for drug discovery requires addressing fundamental challenges in data encoding, feature selection, and algorithm design [209]. Researchers are exploring various approaches to compress molecular descriptors for quantum computing while preserving essential information for prediction tasks.

7.2 Federated learning

Federated learning offers significant potential for addressing privacy and data sharing challenges in biomedical research. This approach enables collaborative training of AI models across multiple institutions without requiring centralized data storage, thereby preserving patient privacy while enabling large-scale collaborative research [210]. Applications of federated learning in healthcare include patient similarity learning, phenotyping, and predictive modeling across multiple medical institutions. The technology enables healthcare organizations to benefit from collective knowledge while maintaining control over sensitive patient data. Privacy-preserving federated learning approaches that incorporate differential privacy provide additional security guarantees while maintaining model performance [211]. These methods enable robust collaborative research while addressing critical concerns about data privacy and security in healthcare applications.

7.3 Digital twins

Digital twins represent an emerging paradigm in personalized medicine, offering the potential to create dynamic, data-driven replicas of individual patients. These virtual models integrate diverse data types, including genetic information, medical imaging, wearable device data, and electronic health records [212]. Digital twins enable simulation of disease progression, optimization of diagnostics, and personalization of treatment plans based on individual genetic and lifestyle profiles. The technology shows particular promise in oncology, where digital twins can simulate tumor responses to different treatment regimens before clinical administration [213]. The integration of AI, IoT, and machine learning technologies with digital twin frameworks enables real-time monitoring, predictive analytics, and dynamic treatment optimization. However, challenges remain in data integration, computational requirements, and ensuring model accuracy across diverse patient populations.

7.4 Blockchain technology

Blockchain technology offers potential solutions for addressing data integrity, security, and sharing challenges in bioinformatics research. The decentralized, immutable nature of blockchain systems can ensure data integrity while enabling secure collaborative research [214]. Applications of blockchain in bioinformatics include secure sharing of genomic data, maintaining research provenance, and enabling transparent collaboration among researchers. Smart contracts can automate data usage agreements and ensure appropriate compensation for data contributions [215]. Blockchain-based systems can address critical challenges in research reproducibility by providing tamper-proof recording of research processes and data. This capability supports the development of verification processes and ensures better guarantees of authenticity and reusability in scientific research [216].

7.5 Explainable AI

Explainable AI (XAI) represents a critical requirement for the adoption of AI technologies in healthcare and drug discovery applications [217]. The need for transparency and interpretability in AI models is particularly important in medical applications where decisions can significantly impact patient outcomes. XAI approaches in bioinformatics include methods such as Shapley Additive Explanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME), which provide insights into model decision-making processes [218]. These methods enable researchers and clinicians to understand and validate AI predictions, supporting trust and adoption of AI technologies. The development of interpretable AI models for omics data analysis has shown significant potential for uncovering biological insights and supporting clinical decision-making [219]. However, challenges remain in balancing model accuracy with interpretability, particularly for complex deep learning architectures.

8 Challenges and limitations

8.1 Data quality and bias

Data quality represents one of the most significant challenges in AI-driven bioinformatics and drug discovery applications. Biological datasets often suffer from issues including incomplete annotations, experimental variability, batch effects, and systematic biases that can significantly impact model performance and generalizability [220].

8.2 Algorithmic bias

Algorithmic bias represents a critical concern in AI applications, particularly when models are trained on datasets that do not adequately represent diverse populations. These biases can lead to disparities in healthcare outcomes and reduced effectiveness of AI systems across different demographic groups [221]. Furthermore, Uncertainty Quantification (UQ) is a critical technique for identifying and mitigating algorithmic bias and model overconfidence. UQ methods, such as Bayesian deep learning or ensemble techniques, provide a measure of how certain a model is about its predictions. This helps define the Applicability Domain (AD) the chemical or biological space where the model's predictions are reliable. By flagging predictions made for molecules or patient populations outside the AD (where uncertainty is high), researchers can prevent biased or erroneous outcomes and make more informed, trustworthy decisions. Addressing these challenges requires comprehensive approaches including improved data collection protocols, standardization of experimental procedures, and development of robust preprocessing and normalization methods [222]. The integration of diverse datasets, careful validation across multiple populations, and the application of UQ to understand model limitations are all essential for developing reliable and equitable AI models.

8.3 Interpretability

The black box nature of many AI models represents a significant barrier to adoption in healthcare and drug discovery applications. Complex deep learning models often provide accurate predictions but lack transparency in their decision-making processes, making it difficult for researchers and clinicians to understand and validate model outputs [223]. Developing interpretable AI models requires balancing accuracy with explainability, often involving trade-offs between model complexity and transparency [224]. Research in explainable AI is addressing these challenges through development of post-hoc explanation methods and inherently interpretable model architectures.

8.4 Computational resources, reproducibility and benchmarking

The computational demands of modern AI applications in bioinformatics are substantial, requiring access to high-performance computing resources and specialized hardware such as GPUs and TPUs [225]. Large-scale molecular simulations, deep learning model training, and emerging quantum computing applications all require significant computational infrastructure. These resource limitations can restrict access to advanced AI technologies, particularly for smaller research institutions and organizations in developing countries, potentially creating an inequitable landscape for innovation [226]. Cloud computing platforms and collaborative computing initiatives are helping to address these challenges by providing shared access to computational resources.

Beyond raw compute power, a critical and related challenge is the reproducibility and benchmarking of AI models. The field often lacks standardized, large-scale benchmarks and rigid evaluation protocols, making it difficult to compare the performance of different algorithms fairly and consistently. Factors such as data leakage, where information from the test set inadvertently influences the training process, the use of different data splitting strategies, and varying evaluation metrics can lead to inflated performance claims that do not hold up in independent validation or real-world settings. The lack of transparency in model architecture, hyperparameters, and training data further exacerbates this issue. Ensuring reproducibility requires a concerted effort towards open science practices, including sharing code, models, and fully specified training workflows, as well as the adoption of community-agreed-upon benchmarks to provide a reliable measure of progress and utility [230].

8.5 Ethical considerations

Privacy and consent represent fundamental ethical challenges in AI applications to healthcare and bioinformatics. The use of personal genetic and health data for AI model development raises important questions about data ownership, consent management, and potential discrimination based on genetic profiles [227]. Algorithmic fairness and the prevention of discriminatory outcomes represent critical ethical requirements for AI systems in healthcare [228]. Ensuring that AI models perform equitably across diverse populations requires careful attention to dataset composition, model validation, and ongoing monitoring of system performance.

8.6 Regulatory aspects

The regulatory landscape for AI applications in drug discovery and healthcare is rapidly evolving, with agencies worldwide developing new frameworks for evaluating AI-driven medical [229]. Regulatory processes must adapt to address the unique challenges posed by AI systems, including issues of model validation, performance monitoring, and post-market surveillance [230]. The integration of AI technologies into drug discovery pipelines requires careful consideration of regulatory requirements throughout the development process. This includes ensuring that AI models meet appropriate standards for safety, efficacy, and quality while maintaining transparency and auditability.

8.7 Data integration and interoperability

A paramount challenge in AI-driven drug discovery is the integration and interoperability of data from highly heterogeneous biomedical sources. Data is siloed across various domainsincluding genomics from one repository, chemical structures from another, clinical trial results from a third, and real-world evidence from electronic health records each with its own formats, identifiers, and metadata standards. This heterogeneity creates significant friction in building unified views of biological systems or patient populations. Harmonizing this data requires resolving entity disambiguation (e.g., ensuring a compound is consistently identified across databases), standardizing ontological terms (e.g., using GO, MeSH, or MONDO ontologies), and developing complex data fusion techniques. Without robust interoperability frameworks, AI models are trained on fragmented data, limiting their ability to capture the complex, multi-scale interactions that underlie disease and drug response, and ultimately restricting the translational power of the resulting insights.

Revised and Expanded 8.4. Computational Resources and Reproducibility The computational demands of modern AI applications in bioinformatics are substantial, requiring access to high-performance computing resources and specialized hardware like GPUs and TPUs [225]. Large-scale molecular simulations, deep learning model training, and emerging quantum computing applications all require significant infrastructure. These resource limitations can restrict access to advanced AI technologies, particularly for smaller research institutions and in developing countries, potentially creating an inequitable landscape for innovation [226].

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9 Future perspectives

While current AI technologies have profoundly impacted early-stage drug discovery, their full translational potential remains nascent. The next decade will be defined by overcoming existing bottlenecks through the development of more integrated, trustworthy, and clinically-aware AI systems. Key frontiers include privacy-preserving collaborative frameworks, multi-modal data fusion, and the creation of dynamic patient avatars.

9.1 Advancing privacy-preserving and collaborative AI with federated learning

The future of robust AI models hinges on access to diverse, large-scale datasets, which are often siloed across institutions due to privacy and regulatory constraints. Federated learning (FL) is poised to become a foundational paradigm, enabling model training across decentralized data sources without sharing raw, sensitive information [24]. This approach not only mitigates privacy risks but also combats algorithmic bias by incorporating data from more diverse patient populations, thereby enhancing model generalizability. The critical research direction will be to overcome FL's practical challenges, including communication overhead, statistical heterogeneity (non-IID data), and ensuring robust model performance across all participating sites. Future work must integrate FL with advanced privacy-enhancing technologies like differential privacy and homomorphic encryption to provide cryptographic guarantees of data security [25]. Success in this area will foster global collaborations, democratize access to AI tools, and build the large, representative datasets necessary for the next generation of predictive models.

9.2 Towards causal and systems-level understanding via multi-modal AI

The integration of multi-omics data (genomics, transcriptomics, proteomics, metabolomics, epigenomics) is essential to move from correlative patterns to a causal, systems-level understanding of disease biology [26]. Future AI models must evolve beyond analyzing these data types in isolation to true multi-modal integration, where interconnected layers of biological information are processed simultaneously. The challenge lies in the high dimensionality, noise, and complex, non-linear relationships within and between these omics layers. The next wave of AI will leverage architectures like transformer-based models and graph neural networks designed to fuse these disparate data types, uncovering emergent biological properties and identifying robust, multi-factorial biomarkers [27]. The ultimate goal is to create AI systems that can predict drug response and resistance not from a single gene signature, but from the dynamic interplay of genetic predisposition, real-time protein signaling, and metabolic state, thereby enabling truly personalized therapeutic strategies [28].

9.3 The paradigm of AI-augmented discovery and digital twins

The future of drug discovery lies not in fully autonomous AI, but in a synergistic, human-AI partnership. A critical manifestation of this is the development of digital twins virtual, dynamic replicas of a patient's pathophysiology [29]. These models will integrate static genetic data with real-time streams from wearable devices, electronic health records, and continuous molecular monitoring. AI-driven digital twins will enable in silico clinical trials, allowing for the virtual testing of thousands of treatment regimens on a simulated population to optimize trial design and predict outcomes with unprecedented accuracy [30]. This paradigm shift will compress development timelines and reduce attrition by prioritizing the most promising candidates for real-world testing. However, realizing this vision requires solving grand challenges in multi-scale modeling, high-fidelity data assimilation, and the establishment of regulatory frameworks for the validation and use of in silico evidence. The focus will shift from AI as a tool for singular tasks to AI as the core of an integrated, patient-centric discovery and development engine.

10 Conclusion

AI-driven bioinformatics has decisively evolved from a promising experimental tool into a cornerstone of modern drug discovery. The integration of predictive modeling, generative chemistry, and breakthroughs in protein structure prediction like AlphaFold and RFdiffusion is fundamentally rewriting the rules of early-stage R&D, compressing discovery timelines and unlocking new frontiers in molecular design.

However, this review underscores that the full translational potential of AI is critically contingent on overcoming persistent and foundational challenges. The remarkable progress in algorithmic power is currently gated by the quality and bias of the underlying data, the "black-box" nature of complex models that hinders clinical trust and regulatory approval, the immense computational resources that limit equitable access, and the nascent state of ethical and regulatory frameworks needed to ensure fairness and safety. The field's reproducibility and benchmarkability also remain significant concerns, often inflated by a lack of standardized evaluation protocols and data leakage.

Therefore, the path forward demands a paradigm shift from purely technical innovation to a more holistic and collaborative approach. A concerted effort among academia, industry, and regulators is essential to establish rigorous benchmarks for reproducibility, foster the development of explainable AI (XAI), champion the creation of diverse and high-quality datasets, and build harmonized regulatory pathways. By navigating these challenges responsibly, AI promises to not only accelerate discovery cycles but also to elevate clinical success rates and democratize access to novel therapeutics, ultimately heralding a true paradigm shift in how we develop medicines for human health.

Author contributions
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Data availability
No datasets were generated or analysed during the current study.

Declarations

Competing interests
The authors declare no competing interests.

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